

RSA Key Notes

What is RSA?

RSA (Rivest–Shamir–Adleman) is an **asymmetric cryptographic algorithm** used for:

- Encryption
- Decryption
- Digital Signatures
- Secure Key Exchange

Unlike symmetric encryption, RSA uses **two different keys**.

Types of RSA Keys

1. Public Key (🔓)

- Can be shared with anyone.
- Used to encrypt data.
- Used to verify digital signatures.
- Cannot be used to generate a valid signature.

Example:

!

```
Public Key
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqh...
-----END PUBLIC KEY-----
```

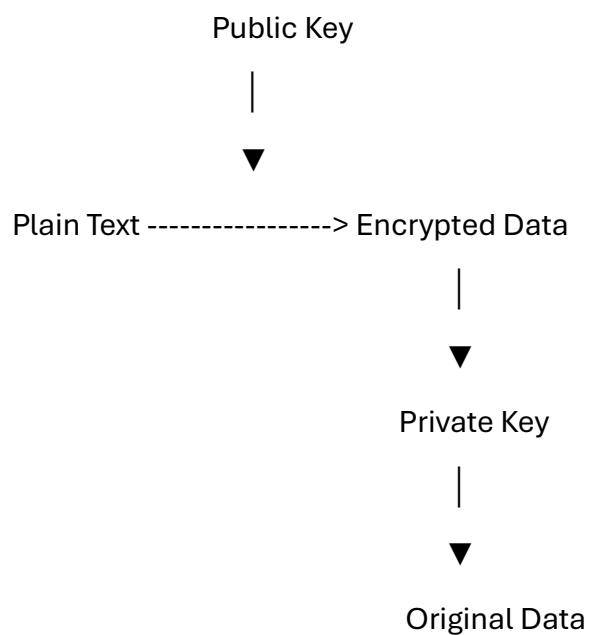
2. Private Key (🔒)

- Must be kept secret.
- Used to decrypt data.
- Used to sign digital signatures.
- Should never be exposed.

Example:

Private Key
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBg...

How RSA Works Encryption Process:



Example:

Alice wants to send confidential information to Bob.

- Bob generates:
 - Public Key
 - Private Key

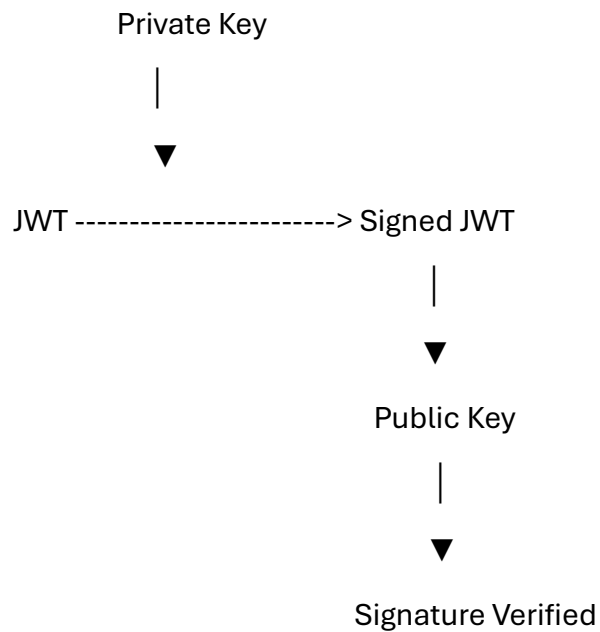
Bob shares only the Public Key.

Alice encrypts the message.

Only Bob can decrypt it using the Private Key.

Digital Signature Process

This is the most common use case in Spring Security.



Here,

Private Key signs.

Public Key verifies.

Why Two Keys?

If only one key existed:

Anyone possessing the key could create fake tokens.

With RSA:

Private Key remains only with the Authentication Server.

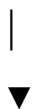
Public Key is shared with all Resource Servers.

Only the Authentication Server can create JWTs.

Every service can verify them.

RSA in Spring Boot (Authentication Flow)

User Login



Authentication Server



Generate JWT



Sign JWT using RSA Private Key



Return JWT



Client



Request with JWT



Resource Server



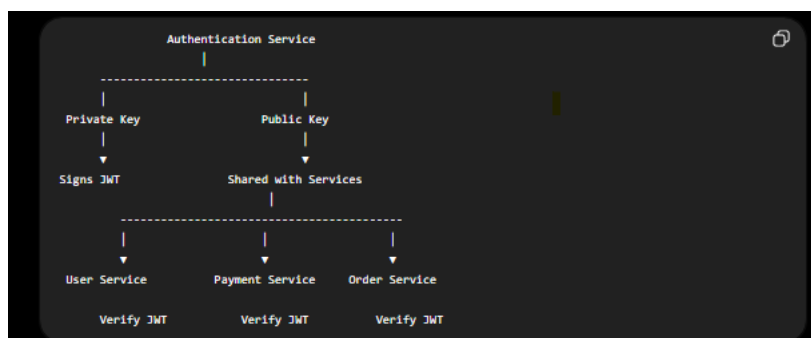
Verify JWT using RSA Public Key



Access Granted

RSA in Microservices

Example Architecture:



Only Authentication Service owns the Private Key.

Other services only verify.

Why RSA is Secure?

Because:

- Public Key cannot derive the Private Key.
 - Only the Private Key can create valid signatures.
 - Public Key can only verify signatures.
 - Prevents attackers from creating fake JWTs.
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Advantages of RSA

- Very secure
 - Uses two-key mechanism
 - No need to share the Private Key
 - Supports Digital Signature
 - Supports Encryption
 - Widely used for JWT Authentication
 - Used in HTTPS certificates
 - Suitable for distributed microservices
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Disadvantages

- Slower than symmetric encryption
 - Not suitable for encrypting large files
 - Larger key sizes increase computational cost
 - Usually used only for key exchange or signatures
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Real-Time Uses

RSA is commonly used in:

- JWT Authentication
- OAuth2
- OpenID Connect
- HTTPS / SSL Certificates
- Banking Applications
- Digital Certificates
- Secure API Communication
- Email Encryption

Interview Questions

1. What is RSA?

RSA is an asymmetric encryption algorithm that uses a Public Key and a Private Key for secure communication and digital signatures.

2. Difference between Public Key and Private Key?

Public Key	Private Key
Shared publicly	Kept secret
Encrypts data	Decrypts data
Verifies signatures	Creates signatures

3. Why is RSA used in JWT?

To digitally sign JWTs using the Private Key while allowing other services to verify them using the Public Key.

4. Can someone generate a JWT using the Public Key?

No.

The Public Key can only verify a JWT.

Only the Private Key can generate a valid digital signature.

5. Why is RSA preferred in Microservices?

Because many services can verify tokens using the Public Key without exposing the Private Key.

6. Is RSA encryption or authentication?

Both.

- Encryption → Encrypt with Public Key, Decrypt with Private Key.
- Authentication/Integrity → Sign with Private Key, Verify with Public Key.

. Easy Trick to Remember

- Encryption
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- Public Key ---> Encrypt
- Private Key ---> Decrypt
- Digital Signature
-
- Private Key ---> Sign
- Public Key ----> Verify

1-Minute Interview Answer

RSA (Rivest–Shamir–Adleman) is an asymmetric cryptographic algorithm that uses a pair of keys: a Public Key and a Private Key. The Public Key can be shared openly, while the Private Key is kept secret. In Spring Security, RSA is primarily used for JWT signing. The Authentication Server signs the JWT using the Private Key, and Resource Servers verify the signature using the Public Key. This ensures that only the Authentication Server can issue valid tokens while allowing multiple microservices to verify them securely without sharing the Private Key. This approach provides authentication, integrity, and trust across distributed systems.